



Tracing source and paleoweathering conditions of the lower jurassic sandstone from sierra de santa rosa section, Northern Sonora, México

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Abstract

The Lower Jurassic sedimentary rocks of the Sierra de Santa Rosa Formation in northern Sonora, Mexico has been deposited in a forearc basin and its age has been assigned based on the fossil assemblages and detrital zircon U–Pb geochronology data. Nevertheless, their geochemical characteristics has not undertaken yet. In this study, the whole rock geochemistry (including major oxides, trace, and rare earth elements) of 23 sandstone samples from the Sierra de Santa Rosa Formation is investigated to infer the provenance and paleoweathering conditions of the basin, within the Lower Jurassic paleogeographic framework. The concentration of total rare earth elements (Σ REEs) range between 46.5 and 166.6 ppm (ave. 118.7 ppm), which is lower than the average Upper Continental Crust (UCC) value (147.80 ppm). Chemical index of weathering (CIA), weathering index of Parker (WIP), and A-CN-K diagrams signify the low to moderate chemical weathering in the source region. It suggests that the source regions experienced arid to semi-arid climatic conditions during the deposition of these detritus. The chondrite normalized REE patterns, negative Eu anomaly, bivariate diagrams and elemental ratios like, Eu/Eu*, La/Sc, La/Co, Th/Sc, Th/Co, and Cr/Th indicate that majority of the detritus were derived from the felsic rocks with minor contribution from intermediate source rocks.

Keywords Provenance · Paleoweathering · Sierra de Santa Rosa Formation · Northern Sonora · México

1 Introduction

The detrital composition of sandstone is largely controlled by the climate, weathering conditions, source rocks, and tectonic setting of the sedimentary basins (Criniti et al., 2023; Critelli et al., 2021; Dickinson & Suczek, 1979; Ingersoll &

Eastmond, 2007; Ramírez-Montoya et al., 2025; Suttner & Dutta, 1986). The major oxides and certain trace elements including high-field strength elements (HFSE), and REE are generally immobile during weathering and transportation, which readily reflect the source rocks composition (Armstrong-Altrin, 2024; Cullers, 1994, 2000; Galindo-Ruiz et

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al., 2018; Nesbitt & Young, 1982, 1984; Ramos-Vázquez & Armstrong-Altrin, 2021; Singh et al., 2022; Sonfack et al., 2021). The geochemistry of the sedimentary rocks was widely used to portray the nature of the source rocks, tectonic setting and the evolution of the continental crust (Armstrong-Altrin et al., 2021, 2022; Cullers et al., 1988; Madhavaraju et al., 2020; Mendita-Lora et al., 2018; Ramachandran et al., 2016; Ramírez-Montoya et al., 2018, 2021; Tawfik et al., 2018; Velmurugan et al., 2019). The concentration of trace elements and their ratios and REE patterns along with Eu anomaly in sediments and sedimentary rocks are commonly used to assess the relative contribution of felsic and mafic sources (Galindo-Ruiz et al., 2025; Madhavaraju, 2015; Madhavaraju et al., 2021, 2024; Olivas et al., 2025; Ramos-Vázquez & Armstrong-Altrin, 2021; Sangeeta et al., 2023; Sawant et al., 2017; Wani & Mondal, 2010).

The Lower Jurassic sedimentary strata are well exposed in the Sierra de Santa Rosa locality (Hardy, 1981). These sedimentary rocks include conglomerate, coarse- to fine-grained sandstone, fossiliferous limestone, siltstone, and mudstone. The stratigraphy, biostratigraphy, and U–Pb geochronological studies were carried out on these strata by several researchers (Damborenea & González-León, 1997; González-León et al., 2005; Hardy, 1981; Hodges et al., 2015, 2017; Pálffy & González-León, 2000; Scholz et al., 2008). An early Sinemurian to middle Toarcian age assigned for the Sierra de Santa Rosa Formation that exposed in the Sierra de Santa Rosa locality (Hodges et al., 2017; Pálffy & González-León, 2000). Previous studies have mainly focused on stratigraphy, paleontology, and geochronology of the sedimentary rocks that exposed in the Sierra de Santa Rosa locality. However, there are still several gaps exist that should be bridged by undertaking geochemical study on these sedimentary rocks. For the present study, geochemical analyses (major, trace, and rare earth elements) were carried out on the sandstone samples collected from the Sierra de Santa Rosa locality to determine the geochemical variations, paleoweathering conditions, and provenance signatures. Also, the results of this study will be compared with the geochemical signatures of the nearby Sierra del Álamo locality to infer regional variations in paleoweathering conditions and source rocks.

2 Geology of the study area

The basement rocks of northwestern Sonora indicate the union of the Mojave, Yavapai, and Mazatzal crustal provinces of southwestern Laurentia (Iriondo et al., 2004; Nourse et al., 2005). The Caborca block in Sonora consists of late Paleoproterozoic rocks of the Mojave and Yavapai provinces, which are intruded by Mesoproterozoic plutons

(Iriondo & Premo, 2010). The Permo-Triassic and Lower Jurassic sedimentary sequence of the El Antimonio Group (~3.4 km-thick) is well exposed in the Sierra del Álamo, ~40 km west of Caborca in northwestern Sonora, México (González-León, 1997; González-León et al., 2000; Lucas et al., 1999) (Fig. 1). El Antimonio Group has been deposited in a forearc basin on the southwestern margin of Laurentia that accumulated marine sediments, and also the sediments were derived from the contemporaneous Permo-Triassic and Jurassic magmatic arcs of North America (González-León et al., 2005; Lindner, 2013). The El Antimonio Group is considered as one of the most complete early Mesozoic sedimentary successions in northwestern México (González-León et al., 2000, 2005) that unconformably overlies the Monos Formation of middle Permian age (Cooper & Arellano, 1946; Dobbs et al., 2021). This group is divided into 16 sequences (sequences I to XVI) (Galindo-Ruiz, 2019; Galindo-Ruiz et al., 2025; González-León et al., 2000, 2005). The age of this group is ranging from the late Permian to early Jurassic (Estep et al., 1997; Galindo-Ruiz et al., 2025; González-León, 1997; González-León et al., 2005; Lucas & Estep, 1999; Lucas et al., 1997; Pálffy & González-León, 2000; Scholz et al., 2008; Szűcs et al., 2019; Taylor et al., 2001).

The Lower Jurassic sedimentary rocks of the Sierra de Santa Rosa Formation show a northwest-to-southeast trend. The two most complete sections of this unit are found in the Sierra de Santa Rosa and Sierra del Álamo localities (Hardy, 1981; Hodges et al., 2017; Galindo-Ruiz et al., 2025; González-León et al., 2005, 2009). Hardy (1973), who first studied the Sierra de Santa Rosa Formation from the locality of the Sierra de Santa Rosa that is located ~100 km to the southeast of the Sierra del Álamo, and he divided it into lower, middle, and upper members (1450 m thick). Based on the ammonoid assemblages (Szűcs et al., 2019; Taylor et al., 2001), the Sierra de Santa Rosa Formation ranges from Upper Hettangian to Upper Sinemurian in the Sierra del Álamo, and from Upper Sinemurian to Lower Pliensbachian in the Sierra de Santa Rosa. However, the detrital zircon U–Pb geochronology reveals Sinemurian to early Toarcian maximum depositional ages for the Sierra del Álamo locality (Galindo-Ruiz et al., 2025) whereas early Sinemurian, Pliensbachian, and middle Toarcian, for the lower, middle, and upper members respectively for the Sierra de Santa Rosa locality (Hodges et al., 2017).

In the study area, the Sierra de Santa Rosa Formation has been divided into lower, middle, and upper members (Hardy, 1981). Subsequently, paleontological and stratigraphic studies were carried out by Damborenea and González-León (1997). They were reported detailed stratigraphy based on the three measured sections, with total thickness of 843 m (Fig. 2). The lower member includes conglomerate,

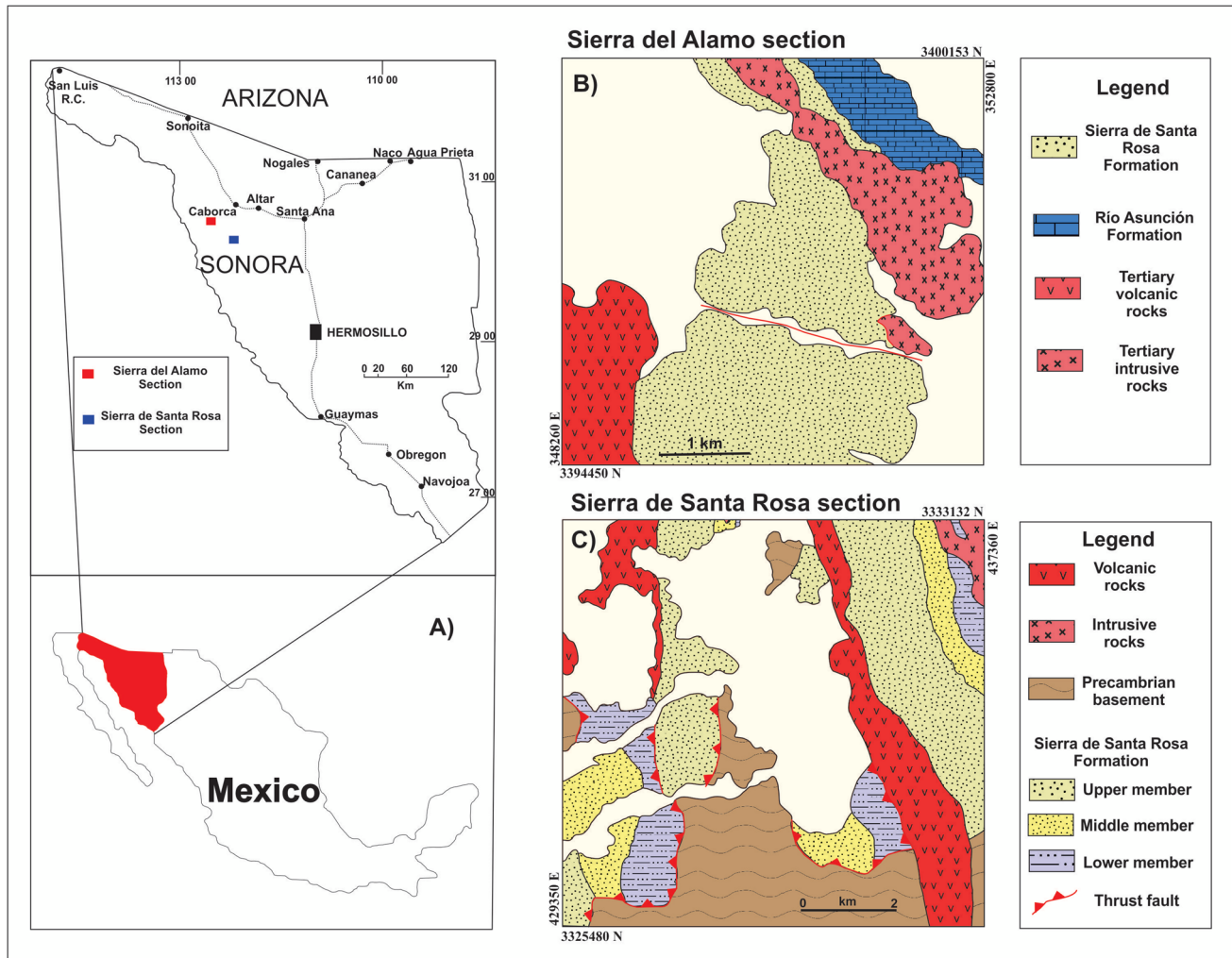


Fig. 1 A) Location map of Sierra de Santa Rosa and Sierra del Álamo section, Sonora, México; B) Geological map of the Sierra del Álamo area (modified from Galindo-Ruiz et al., 2025) C) Geological map of the Sierra de Santa Rosa area (modified from Hodges et al., 2017)

coarse- to fine-grained sandstone, siltstone, and mudstone that were deposited under fluvial environment (Damborenea & González-León, 1997). It contains bivalves like *Weyla alata* von Buch (Damborenea & González-León, 1997), *Weyla bodenbenderi*, *Frenguelliella poultoni*, Lucinidae gen. et sp. indet A, and Lucinidae gen. et sp. indet B (Scholz et al., 2008). It also contains ammonoid *Paltechioceras* that indicate the uppermost Sinemurian Harbledownense Zone (Pálffy & González-León, 2000).

The middle member consists of fossiliferous limestone interbedded with mudstone and sandstone and it contains numerous bivalve taxa including *Weyla alata*, *Weyla bodenbenderi*, *Weyla (Lywea) unca*, and *Weyla titan* and colonial corals (Damborenea & González-León, 1997; González-León et al., 2005; Hodges et al., 2015; Scholz et al., 2008). The upper member consists of calcareous, fine- to medium-grained sandstone, calcareous mudstone, limestone with bivalve assemblages, and carbonate buildups

with reef-building corals. It contains ammonite's species like *Metaderoceras* sp., *Dubariceras* sp., and *Fuciniceras* sp. (Pálffy & González-León, 2000). The evidence of volcanic activity with well-developed tuff beds are seen in the middle part of the Sierra de Santa Rosa Formation in the Sierra del Álamo locality, whereas it is absent in the Sierra de Santa Rosa locality. Similarly, ammonites and their zonations were well reported from the Sierra del Álamo locality, whereas it is scarce in the Sierra de Santa Rosa locality.

3 Methods and materials

3.1 Geochemical analysis

Twenty-three sandstone samples were selected for geochemical study. For major oxides, the powdered samples were fused in an M4 automatic fluxer by Claisse with a 10%

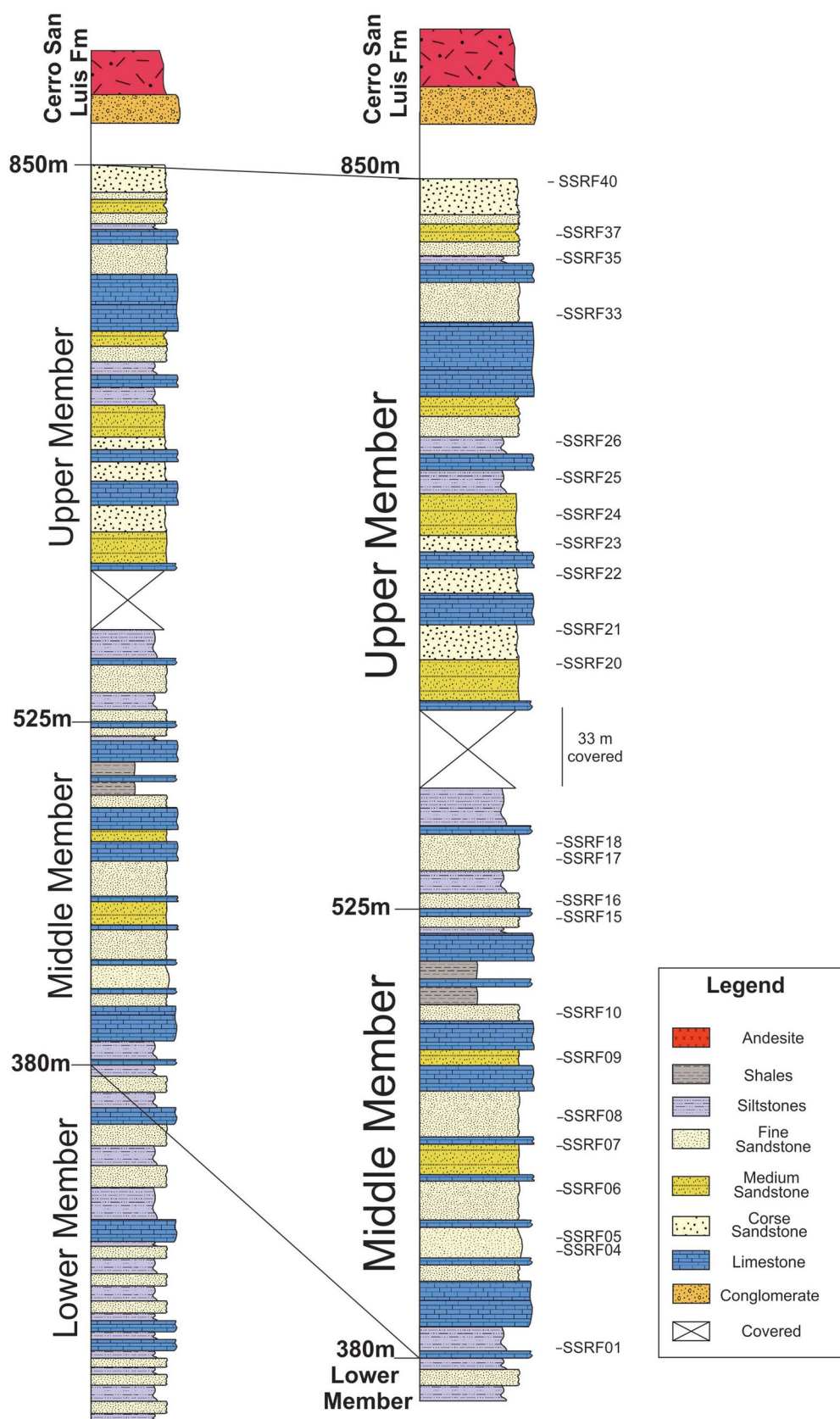


Fig. 2 Stratigraphic section of the Sierra de Santa Rosa Formation in the Sierra de Santa Rosa locality, Sonora, México (after Damborenea & González-León, 1997; Hodges et al., 2017)

dilution in $\text{LiBO}_2/\text{Li}_2\text{B}_4\text{O}_7$ 1:1 flux mix adding 50 μl of LiI solution with 250 g/l. The beads were analyzed using X-ray fluorescence spectroscopy (Rigaku Primus II) at Instituto de Geología, Universidad Nacional Autónoma de México (UNAM). The IGLa-1 internal geochemical standard was used for data quality control (Lozano & Bernal, 2005). The analytical precision was better than $\pm 3\%$ for SiO_2 , Fe_2O_3 , CaO , MgO , MnO , and P_2O_5 and better than $\pm 5\%$ for Al_2O_3 , Na_2O , K_2O and TiO_2 . One gram of sample was heated at 950 °C in porcelain crucibles for 2 h to measure loss on ignition (LOI). Whole rock trace elements, including rare earth elements were determined using a Thermo ICap QcICPMS at Laboratorio de Estudios Isotópicos, Instituto de Geociencias (CGEO), UNAM, according to analytical procedures described by Mori et al., (2009). The International geochemical standards AGV-2, BCR-2, JB2 and BHVO-2 were used to monitor the data quality. The analytical precision errors for trace and rare earth elements were better than $\pm 4\%$. The chondrite data proposed by Taylor and McLennan (1985) were used for normalization of REEs.

3.2 Geochemical proxies

The Chemical Index of Alteration (CIA) serves as an important parameter for assessing weathering intensity, defined as $\text{CIA} = \text{Al}_2\text{O}_3 / (\text{Al}_2\text{O}_3 + \text{CaO}^* + \text{Na}_2\text{O} + \text{K}_2\text{O}) \times 100$ (Nesbitt & Young, 1982). Weathering Index of Parker (WIP: Parker, 1970) is another proxy to estimate the chemical weathering intensity of the source rocks. WIP index is calculated as follows: $\text{WIP} = [(\text{CaO}^*/0.7 + 2 \text{Na}_2\text{O}/0.35 + 2 \text{K}_2\text{O}/0.25 + \text{MgO}/0.9)] \times 100$. Both CIA and WIP indices are calculated using molar percentages of oxides, with CaO^* referring as CaO present in silicate fractions alone. The CaO^* correction method is detailed by McLennan et al. (1993) and Bock et al. (1998) in which CaO values were accepted only if $\text{CaO} < \text{Na}_2\text{O}$, when $\text{CaO} > \text{Na}_2\text{O}$, it was assumed that the concentration of CaO is equal to Na_2O . In addition, the Al_2O_3 - ($\text{CaO}^* + \text{Na}_2\text{O}$) - K_2O (A-CN-K) ternary plot is considered as powerful tool to evaluate the weathering trends than a simple way of comparing numerical values (Nesbitt & Young, 1984).

4 Results

4.1 Elemental variations

The major oxides concentrations of sandstone samples of the Sierra de Santa Rosa Formation are given in Supplementary Table 1. The concentrations of SiO_2 , Al_2O_3 and CaO are higher in the sandstone samples (46.6 to 84.5 wt.%, 5.9 to 15.3 wt.%, and 0.37 to 22.6 wt.%, respectively) than other

oxides. The higher content of CaO at certain intervals of the studied section is related to the enrichment of carbonate minerals. The sandstone samples have relatively higher concentrations of Fe_2O_3 and MgO (0.79 to 7.17%; 0.40 to 3.71%; respectively) than other elements. Large variations are observed in K_2O and Na_2O contents (0.69 to 4.67%; 0.73 to 5.19%; respectively). Some samples are enriched in Na_2O than K_2O (Supplementary Table 1). The higher concentration of Na_2O may be due to the presence of Na-plagioclase and other Na-bearing minerals. The concentrations of MnO , TiO_2 , and P_2O_5 are generally low in the studied samples (Supplementary Table 1), except a sandstone sample that show higher value of TiO_2 (SSRF 20: 1.17%). The average values of the sandstone samples are compared with average UCC values (Rudnick & Gao, 2003), indicate that SiO_2 content is similar, whereas CaO content is enriched, while Al_2O_3 , Fe_2O_3 , MgO , K_2O , Na_2O , TiO_2 , and P_2O_5 contents are significantly depleted (Supplementary Table 1). SiO_2 indicates significant negative correlation with MnO , CaO , and P_2O_5 ($r = -0.74$, -0.75 , -0.62 , respectively; Supplementary Table 2). TiO_2 have significant positive correlation with Al_2O_3 ($r = 0.65$; Supplementary Table 2) indicates a significant concentration of rutile in the sandstone (Khan et al., 2025). A positive correlation is observed between Al_2O_3 and Fe_2O_3 , MgO , K_2O , and TiO_2 supports their association with aluminosilicates ($r = 0.78$, 0.53 , 0.63 , 0.65 , respectively; Supplementary Table 2).

The concentration of trace elements in sandstone from the Sierra de Santa Rosa Formation are listed in Supplementary Table 3. The trace elements like Nb, Y, Sc, Cr, Th, and Zr are more useful to interpret the source rock characteristics because they are less affected by weathering and diagenesis (Cullers, 2000). The content of Ba is comparatively higher than other trace elements that ranging from 188.6 to 1348.7 ppm, with an average value of 579.5 ppm. The concentration of trace elements like Zr (125.20 to 304.88 ppm, avg. 221.22 ppm), Sr (65.46 to 393.32 ppm, avg. 223.85 ppm), Rb (24.42 to 157.73 ppm, avg. 87.20 ppm), Zn (19.28 to 142.30 ppm, avg. 55.74 ppm), V (18.97 to 146.71 ppm, avg. 56.84 ppm), and Cr (7.70 to 44.77 ppm, avg. 20.55 ppm) are varied significantly. The remaining trace element concentrations are generally low < 5 ppm (Supplementary Table 3). The average values of most of the trace elements are depleted when compared to UCC (Rudnick & Gao, 2014) except Rb, Y, Zr, and Y (Supplementary Table 3). Both Rb and Ba have significant positive correlation with K_2O ($r=0.84$, $r=0.61$, respectively) suggests that their distribution is controlled by clay minerals. The high field strength elements like Nb, Y, Th, and U also have positive correlation with K_2O ($r=0.60$, $r=0.52$, $r=0.71$, $r=0.53$, respectively) indicating that these elements are mainly hosted in clay minerals. Similarly, a strong

correlation is observed between Sr and CaO ($r=0.74$) reflect that this element is largely controlled by carbonate minerals.

The concentrations of REE and their ratios are given in Supplementary Table 4. Σ REE content in sandstone samples are ranging from 46.5 to 166.6 ppm with an average of 118.7 ppm. The mean Σ REE value is lower than the UCC value (147.8 ppm). Few sandstone samples (SSRF-06, SSRF-07, and SSRF-08) have higher concentration of SiO_2 (Supplementary Table 1) than others, consequently their Σ REE contents are also low (Supplementary Table 4). Such

low values in Σ REE contents could be possibly due to dilution effect of quartz (Cullers, 2000). The REE distribution pattern in the sandstone is mostly uniform, exhibits LREE enriched, relatively flat pattern with little fractionation (La_N/Yb_N : 4.98 – 9.97; Fig. 3). Most of the sandstone samples show negative Eu anomaly (Eu/Eu^* : 0.54 – 0.93, $n=22$), except a sandstone (SSRF-16) show significant positive Eu anomaly (Eu/Eu^* : 1.57). Moderate to strong positive correlation observed between Σ REE and certain major oxides (Al_2O_3 , Fe_2O_3 , MgO , K_2O , and P_2O_5) and trace elements

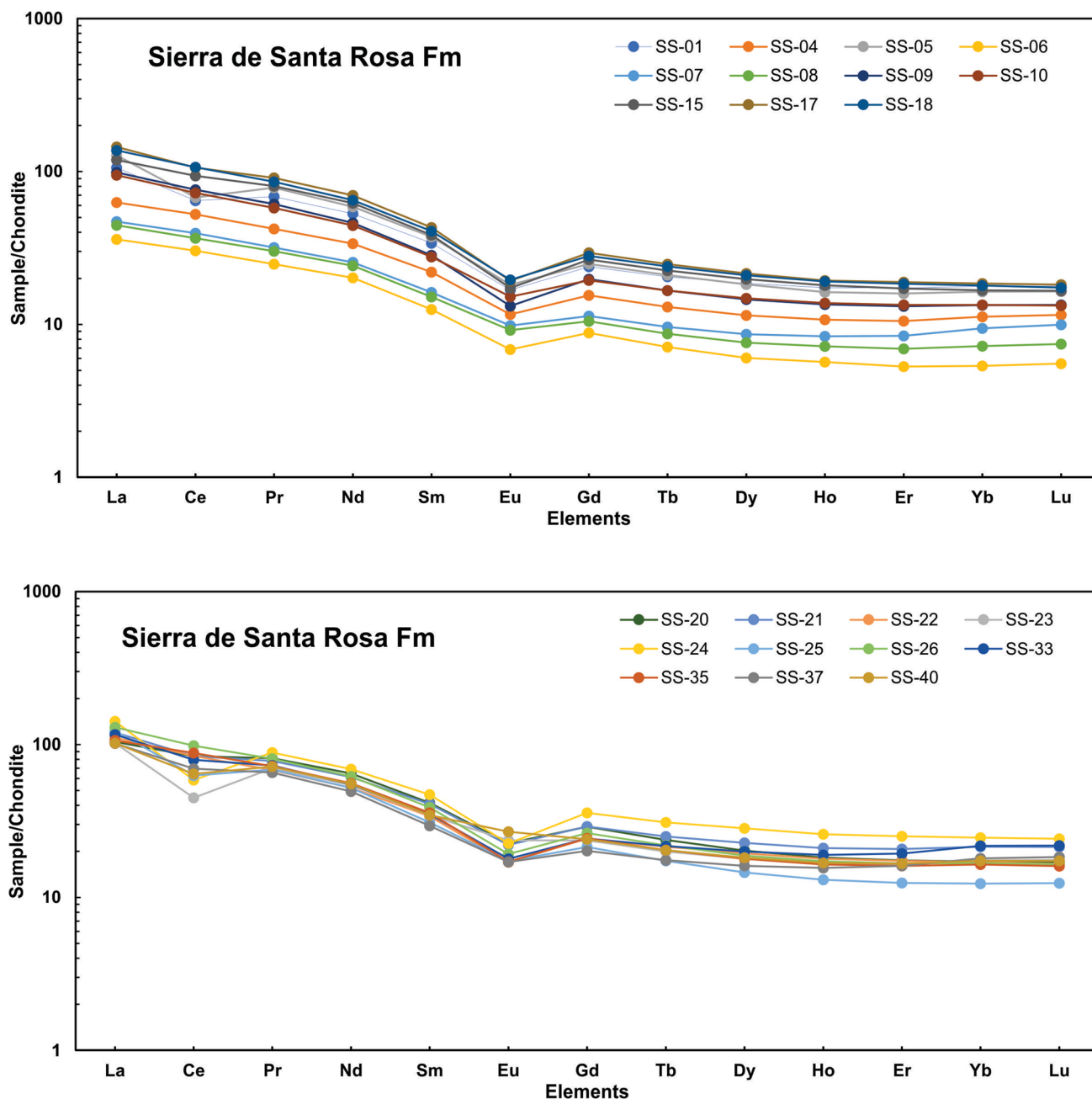


Fig. 3 Chondrite-normalized rare-earth elements patterns for sandstone samples of the Sierra de Santa Rosa section, Sonora, México (chondrite normalized values after McDonough & Sun, 1995)

(Sc, Zn, Ga, Rb, Y, Nb, Ba, Pb, Th, and U) suggests that the accessory minerals and aluminosilicates (mainly mica and clay minerals) may control the distribution of rare earth elements in the sandstone (McLennan, 1989).

5 Discussion

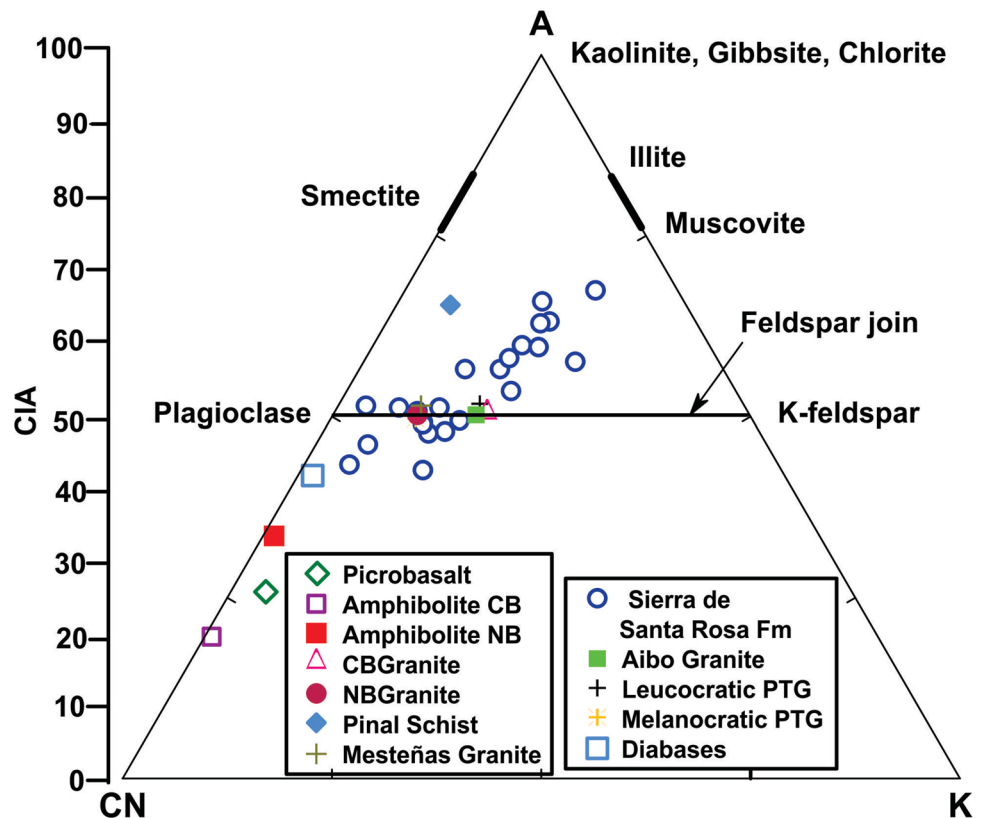
5.1 Paleoweathering

Composition of sedimentary rocks reflects the intensity of source-area weathering, because major elements exhibit quite different geochemical behaviors during weathering processes (McLennan, 1989). The intensity of chemical weathering is calculated by using several indices like chemical index of alteration (CIA: Nesbitt & Young, 1982), weathering index of Parker (WIP: Parker, 1970), chemical index of weathering (CIW: Harnois, 1988), and plagioclase index of alteration (PIA: Fedo et al., 1995). Among them, the widely used index to understand the paleoweathering conditions of source area is CIA. The fresh rocks have CIA values around 50 whereas the intensively weathered products such as kaolinite and gibbsite have values up to 100 (Nesbitt & Young, 1982). The CIA values for the sandstone range between 43 and 67 (54 ± 6.8 , $n=23$). Two sandstone samples from the lower part of the section (SSRF-04: 46; SSRF-06: 43) and one sample from the upper part of the

section (SSRF-37: 43) have lower CIA values suggesting that they contributed by unweathered source rocks. However, the remaining sandstone samples have significant variations in CIA values (48 to 67) suggesting that they were received sediments from low to moderately weathered rocks in the source areas.

The CIA values are plotted on the A-CN-K diagram to infer the weathering trend and K-metasomatism (Fedo et al., 1995; Nesbitt & Young, 1982, 1984), where A, CN, and K represent the molar fractions of Al_2O_3 , $\text{CaO} + \text{Na}_2\text{O}$, and K_2O , respectively. The weathering trend of parent rocks is subparallel to the A-CN apex on A-CN-K diagram (Nesbitt & Young, 1984) whereas sediments affected by K-metasomatism mainly plot towards K-axis (Fedo et al., 1995). On the A-CN-K diagram (Fig. 4), most of the sandstone samples plot near and above the feldspar join line and also plot subparallel to A-CN axis with moderate dispersion suggesting a low to moderate degree of chemical weathering. This is consistent with the CIA values (Supplementary Table 1). However, few samples plot just below the feldspar join line which is consistent with their low CIA values ((SSRF-04: 46; SSRF-06: 43; SSRF-37: 43). The CIA values of these samples are lower than the average UCC value (UCC: 50) that may likely due to unweathered nature of the source rocks or enrichment of feldspar in the coarse fractions due to sedimentary sorting process.

Fig. 4 A-CN-K (molar proportion) ternary diagram (Nesbitt & Young, 1984) for sandstone samples of the Sierra de Santa Rosa section, Sonora, México



Another index of weathering (W index) and M-F-W diagram was proposed by Ohta and Arai (2007), where M, F, and W apexes indicate the mafic rock, felsic rock, and degree of weathering fields, respectively. On the M-F-W diagram (Fig. 5), the sandstone samples plot in the felsic field and also, they trend towards W apex with greater dispersion suggesting a low to moderate chemical weathering condition. In addition, trace elements such as Rb, and Sr and their ratios are helpful to infer the extent of paleoweathering in the source region (McLennan et al., 1993). The sandstone samples show wide range in Rb/Sr ratio (0.12 to 2.34) with an average of 0.54 ($n=23$) which is higher than UCC value (0.32) and lower than average of PAAS (0.80; Taylor & McLennan, 1985) suggest that the source area has suffered a low to moderate intensity of chemical weathering.

The CIA and WIP indices are mostly used by various researchers to evaluate the weathering history and climatic conditions of the source rocks (Armstrong-Altrin et al., 2013; Galindo-Ruiz et al., 2025; Madhavaraju et al., 2016a, 2016b, 2019; Olivas et al., 2025; Ramírez-Montoya et al., 2021; Wani & Mondal, 2010). The usage of CIA vs WIP plot readily discriminate between first cycle sediments and polycyclic sediments (Garzanti & Resentini, 2016; Garzanti et al., 2013, 2014) and also discriminates the sorting and recycling effects (Guo et al., 2018). The first cycle sediments have CIA/WIP ratios between 1 and 2 and show negative correlation between CIA and WIP (Garzanti et al., 2014). On the CIA vs WIP plot (Fig. 6), most of the

sandstone samples show inverse relationship with limited dispersion indicates that they were largely contributed by first-cycle weathering products. However, few samples plot near and below the carbonate dilution line suggesting that these samples were some extent affected by carbonate dilution process.

5.2 Sedimentary sorting and recycling

The immobile elements like Al, Ti, and Zr are generally enriched during chemical weathering and are considered as good proxies for sorting and recycling (Kairouani et al., 2023; Perri et al., 2021). During sedimentary sorting, SiO_2 and Al_2O_3 may fractionate from quartz and feldspar and clay minerals, respectively, whereas TiO_2 is present within Fe/Ti hydroxides or clay minerals. Similarly, the enrichment of Zr in the sedimentary rocks is mainly due to the sorting and recycling effects (Caracciolo et al., 2011; García et al., 1994; Ramos-Vázquez et al., 2018). The sorting and recycling processes can be inferred by using Al_2O_3 -Zr-TiO₂ ternary diagrams (Caracciolo et al., 2011; García et al., 1994). On this diagram (Fig. 7), sandstone samples plot close to the granodiorite suggesting the immature nature of the studied samples with poor sorting and recycling effects. In addition, the recycling process can be defined by using Th/Sc vs Zr/Sc plot (McLennan et al., 1993). On this diagram, sandstone samples plot along the compositional variation line

Fig. 5 MFW diagram (Ohta & Arai, 2007) shows weathering trend for sandstone samples of the Sierra de Santa Rosa section, Sonora, México

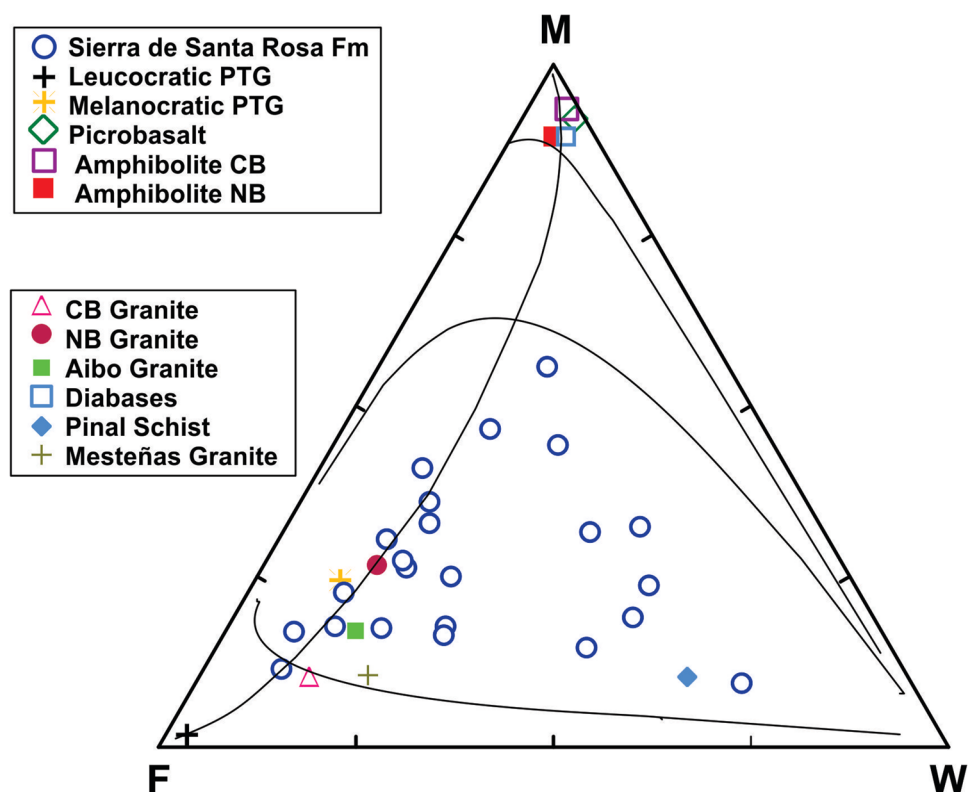


Fig. 6 WIP vs CIA bivariate plot (Garzanti et al., 2014) shows weathering trend and carbonate dilution effect for sandstone samples of the sierra de santa rosa section, Sonora, México

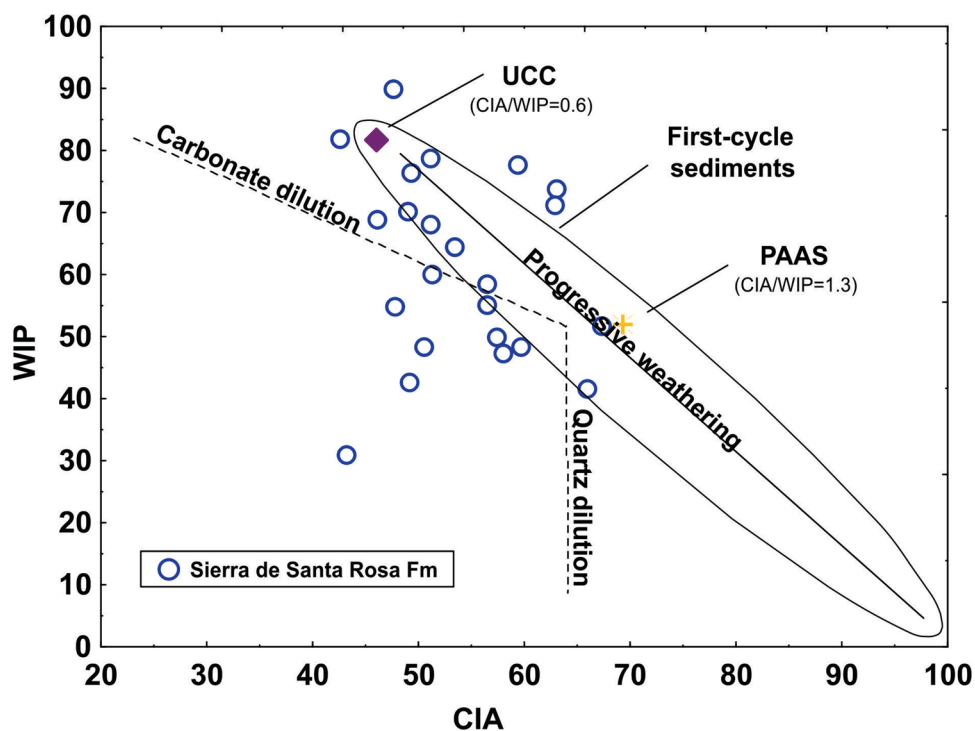
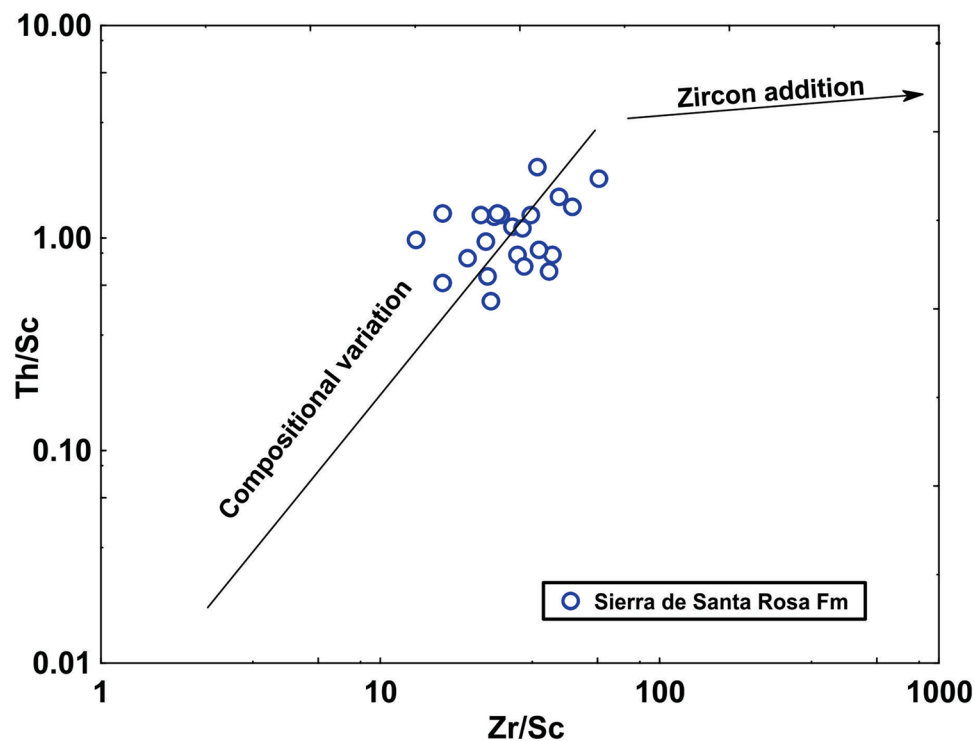


Fig. 7 Th/Sc vs Zr/Sc bivariate plot (McLennan et al., 1993) for sandstone samples of the Sierra de Santa Rosa section, Sonora, México. Trend 1 shows compositional variations, whereas trend 2 indicates zircon addition due to sediment sorting and recycling



suggesting that these sediments were not affected by sorting and recycling processes.

5.3 Provenance

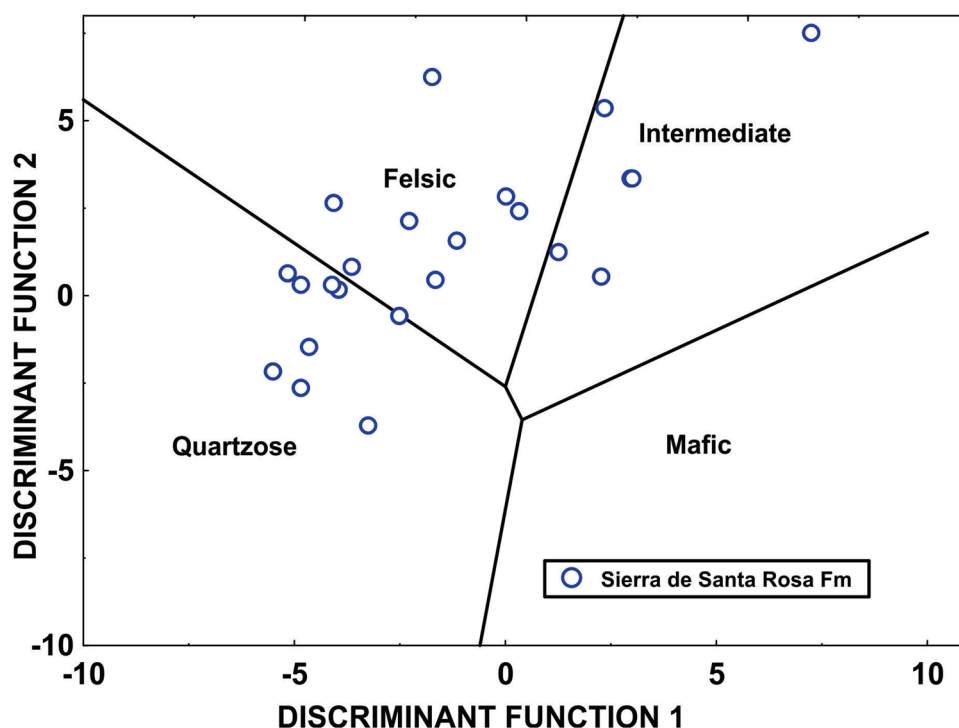
The provenance of the sedimentary rocks can be assessed using major oxides concentrations, particularly Al_2O_3 and TiO_2 are more useful because Al and Ti are immobile during transportation and weathering (McLennan et al., 1990). Hence, $\text{Al}_2\text{O}_3/\text{TiO}_2$ ratio is commonly used to estimate the nature of the source rocks (Garcia et al., 1994; Hayashi et al., 1997; Olivas et al., 2025; Ramírez-Montoya et al., 2021). $\text{Al}_2\text{O}_3/\text{TiO}_2$ ratios significantly vary in mafic, intermediate, and felsic rocks (3 to 8, 8 to 21, 21 to 70, respectively; Supplementary Table 1). The sandstone samples have a wide range of $\text{Al}_2\text{O}_3/\text{TiO}_2$ ratio (~11–31), which indicates their derivation from both felsic and intermediate source rocks. Major oxides-based discriminant function diagram was proposed by Roser and Korsch (1988) to infer source rocks characteristics. The diagram includes four groups: mafic igneous, intermediate igneous, felsic igneous, and quartzose. On this diagram (Fig. 8), most of the sandstone samples plotted in the felsic and intermediate fields. However, some samples fall in the quartzose region indicating loss of feldspar and increase in quartz due to weathering.

The geochemical elements in the sedimentary rocks provide lithological characteristics of the source rock, particularly REE, because they are least soluble, making them relatively stable and unaffected by transportation and depositional processes (Taylor & McLennan, 1985). The

sandstone samples of the Sierra de Santa Rosa Formation have LREE enriched and virtually flat HREE in the chondrite-normalized REE patterns and this characteristic feature indicate that the sediments were probably derived from the felsic source rocks. La/Th against Hf diagram is useful to discriminate between various source rocks such as felsic, mixed felsic/mafic, andesitic arc, tholeiitic oceanic arc, and recycled sources (Floyd & Leveridge, 1987). Low La/Th ratio signifies felsic source whereas higher ratio indicates tholeiitic oceanic arc source. Similarly, lower content of Hf suggests the felsic source, whereas increasing concentration of Hf indicates the recycled nature (Floyd & Leveridge, 1987). On this diagram (Fig. 9), majority of the sandstone samples plot in the felsic source region and also plot closer to the Precambrian granites of the Caborca and North American blocks (Irriondo et al., 2004). However, few samples fall away from the felsic source and plot along the recycled source. In addition, the source rocks characteristics of the siliciclastic rocks can also be differentiated using the La/Sc vs Co/Th diagram (McLennan et al., 1993). The cross plot reveals that the relationship between La/Sc and Co/Th ratios show higher values of La/Sc compared to Co/Th indicating felsic source (Fig. 10). Further, most of the samples plot near to lueco and melanocratic Permo-Triassic granites (Arvizu et al., 2009), Pinal Schist (Madhavaraju et al., 2017), Caborca block granite, and North American block granite (Irriondo et al., 2004).

Certain trace and rare earth elements such as Zr, Hf, Nb, Y, Th, U, and La are preferentially separated into melts during crystallization (Feng & Kerrich, 1990) and these elements

Fig. 8 Provenance discriminant diagram for sandstone samples of the Sierra de Santa Rosa section, Sonora, México (after Roser & Korsch, 1988)



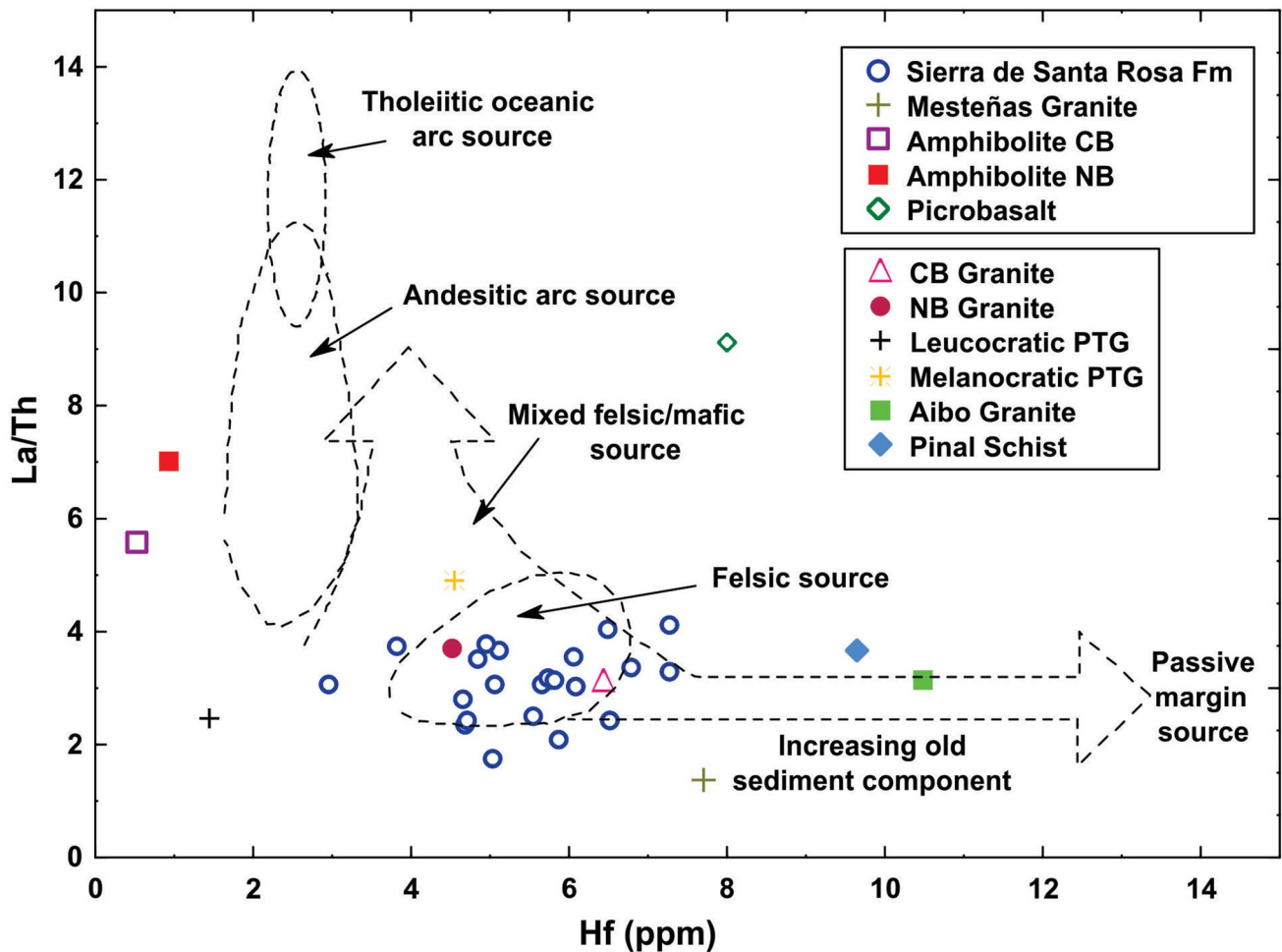


Fig. 9 La/Th vs Hf bivariate plot for sandstone samples of the Sierra de Santa Rosa section, Sonora, México (Floyd & Leveridge, 1987)

are particularly enriched in felsic rocks (Lee, 2002). However, compatible elements (Sc, Cr, Co, Ni, and V) are readily concentrated in mafic rather than felsic rocks (Rollinson, 1993). The elemental ratios like, La/Sc, La/Co, Th/Sc, Th/Co, Cr/Th, and Eu/Eu* are served as indicators of source rocks (Cullers, 2002; Madhavaraju, 2015; Madhavaraju et al., 2010; Raza et al., 2012). La/Sc (1.58 – 5.98, ave. 3.33), La/Co (1.42 – 14.12, ave. 5.16), Th/Co (0.77 – 4.63, ave. 1.72), and Cr/Th (0.77 – 5.44, ave. 2.86) ratios have wide variations where as Th/Sc (0.50 – 2.15, ave. 1.10) shows limited variations. These elemental ratios have been compared with upper continental crust (UCC), post-Archean Australian Shale (PAAS), felsic, and mafic rocks suggest that these ratios are within the range of felsic rocks, however, some samples fall within the range of intermediate source rocks composition.

5.4 Regional variations in paleoweathering and provenance

The siliciclastic rocks of the Sierra de Santa Rosa Formation are well exposed in the localities of Sierra del Alamo and Sierra de Santa Rosa. The geochemical data of the present study (Sierra de Santa Rosa section) is compared with Sierra de Álamo section (Galindo-Ruiz et al., 2025) to infer the regional variations in paleoweathering condition and provenance signatures. The geochronological studies confirm their contemporaneous sedimentation occurred in these localities during the Early Jurassic (Galindo-Ruiz et al., 2025; Hodges et al., 2017).

Although there are a wide variety of indices, the CIA (Nesbitt & Young, 1982) has proven to be a reliable source of information to determine the degree of weathering in ancient rocks (Galindo-Ruiz et al., 2025; Pacheco-Olivas et al., 2025; Ramírez-Montoya et al., 2021). The sandstones of this study show a range of CIA values from 43 to 67 (n=23), suggesting low to moderate chemical weathering

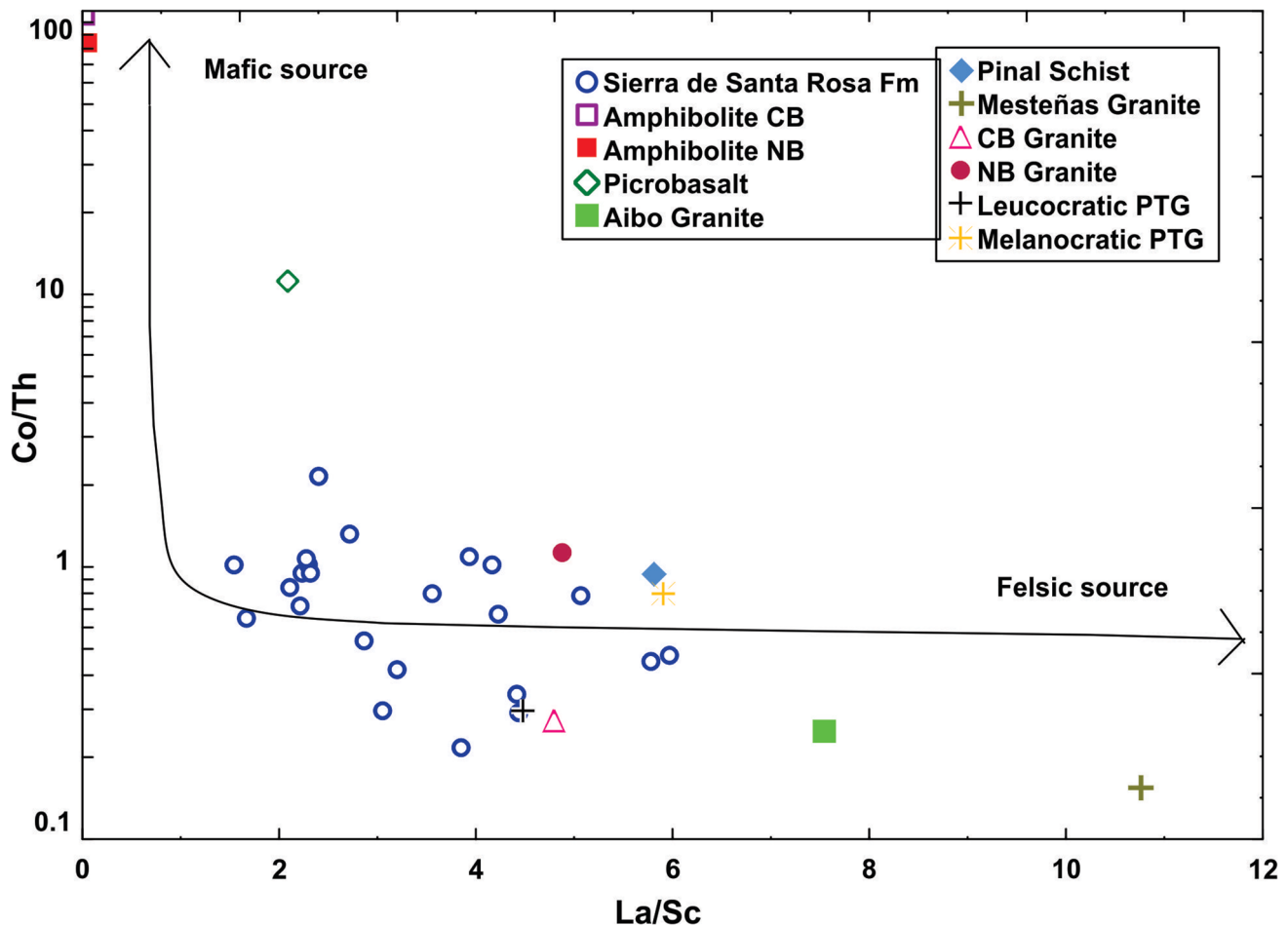


Fig. 10 Co/Th vs La/Sc bivariate plot showing source rock characteristics for sandstone samples of the Sierra de Santa Rosa section, Sonora, México

conditions. Similarly, the rocks from Sierra del Álamo show a similar behavior, with a CIA range of 38 to 76 ($n=31$), also indicating low to moderate chemical weathering (Galindo-Ruiz et al., 2025). CIA values of both units are plotted in the A-CN-K diagram (Nesbitt & Young, 1982, 1989), where both sequences appear to follow a similar trend, however, some variations can be observed (Fig. 11).

Some sandstone samples from this study plot below the feldspar line and majority of the samples plot above the feldspar join line and forms a linear trend along A-CN trend line. However, the samples from the Sierra del Álamo locality are divided into two groups, one group plot below the feldspar line, whereas another group plot above the feldspar join line and show line trend along A-CN line. These CIA values suggest that the sediments were derived from rocks with little chemical weathering. In addition, when comparing these values with the CIA values of source rocks, a strong affinity is observed with the granitic rocks of the Permo-Triassic volcanic arc (Arvizu & Iriondo, 2015; Arvizu et al., 2009; Barth & Wooden, 2006), as well as with granitic rocks

that make up the Proterozoic basement and the rocks that intrude it (Anderson & Silver, 2005; Iriondo et al., 2004; Madhavaraju et al., 2017; Nourse et al., 2005). Further, on CIA vs WIP diagram (Garzanti et al., 2014), samples from both sections suggest a progressive weathering trend; however, several samples follow the trend of quartz dilution effects, suggesting the presence of sedimentary recycling in some samples (Fig. 12). In addition, on the M-F-W diagram (Fig. 13), samples from both sections suggest that source rocks are felsic to intermediate compositions, however, they show some variations in weathering trends.

These models suggest that the chemical weathering conditions in the source areas of both localities were ranging from low to moderate weathering, so these regions experienced mainly arid to semi-arid climatic conditions, which is consistent with the model proposed by Suttner and Dutta (1986) in which almost all the samples fall into the arid and semi-arid fields (Fig. 14). These results are consistent with warm and arid climatic conditions proposed by various authors for the Jurassic rocks in different regions (Chandler

Fig. 11 A-CN-K (molar proportion) ternary diagram (Nesbitt & Young, 1984) showing the variations in chemical weathering of samples collected from Sierra del Álamo and Sierra de Santa Rosa sections of Sierra de Santa Rosa Formation, Sonora, México (Sierra del Álamo data from Galindo-Ruiz et al., 2025)

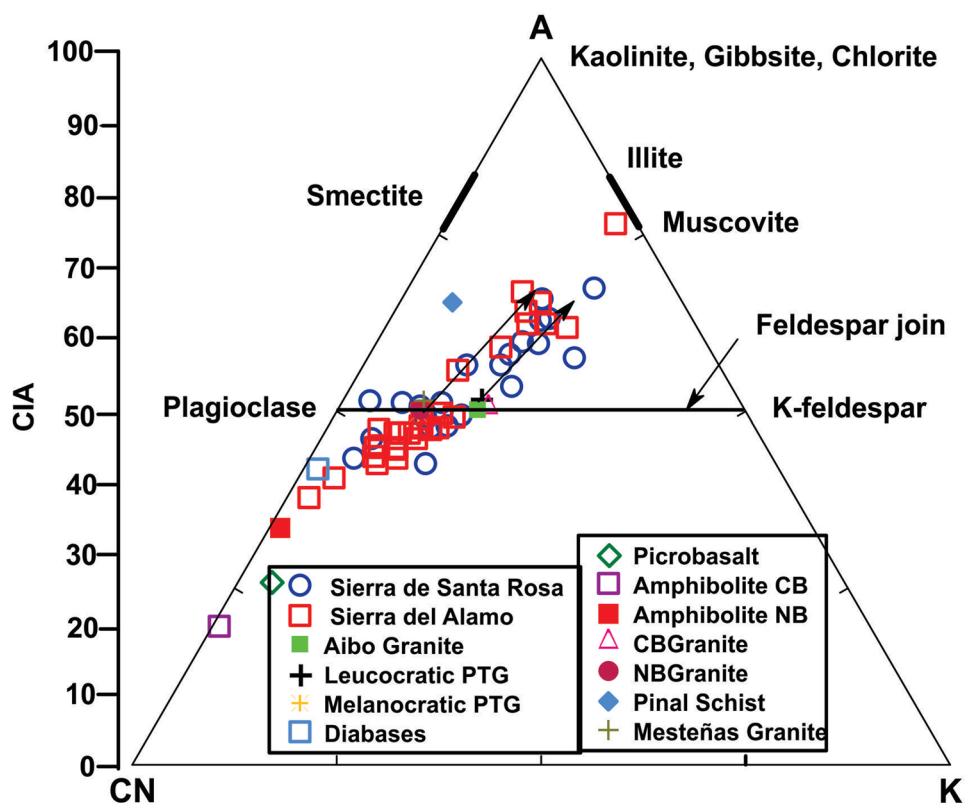


Fig. 12 WIP vs CIA bivariate plot (Garzanti et al., 2014) shows variations in weathering trend, carbonate dilution, and quartz dilution effects for samples collected from Sierra del Álamo and Sierra de Santa Rosa sections of Sierra de Santa Rosa Formation, Sonora, México (Sierra del Álamo data from Galindo-Ruiz et al., 2025)

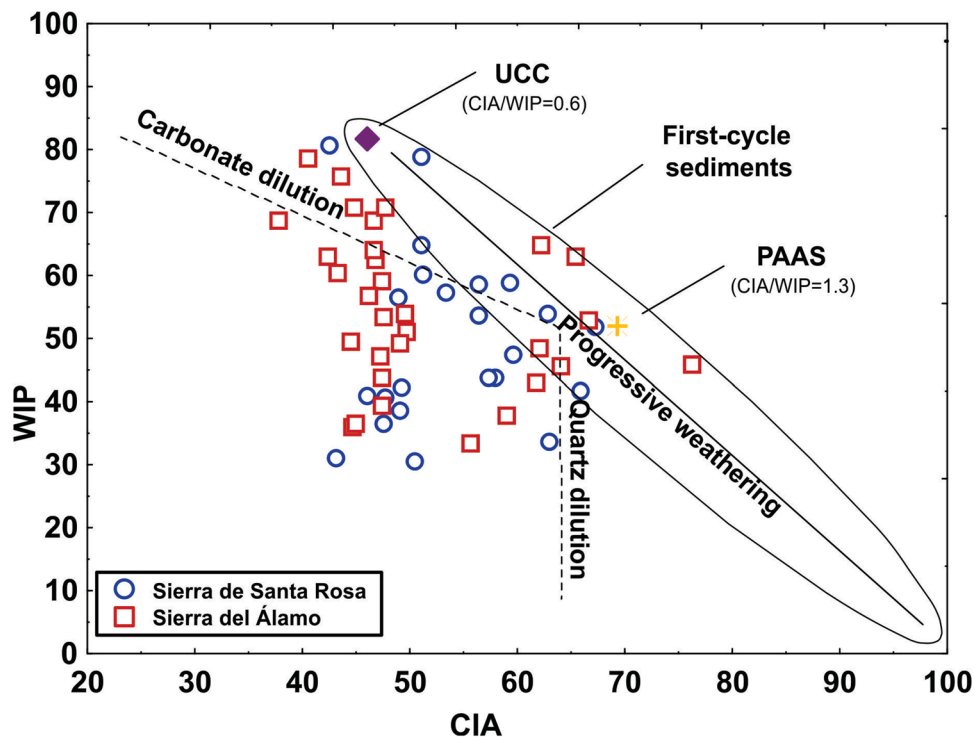


Fig. 13 Variations in weathering trends are showing MFW diagram (Ohta & Arai, 2007) for samples collected from Sierra del Álamo and Sierra de Santa Rosa sections of Sierra de Santa Rosa Formation, Sonora, México (Sierra del Álamo data from Galindo-Ruiz et al., 2025)

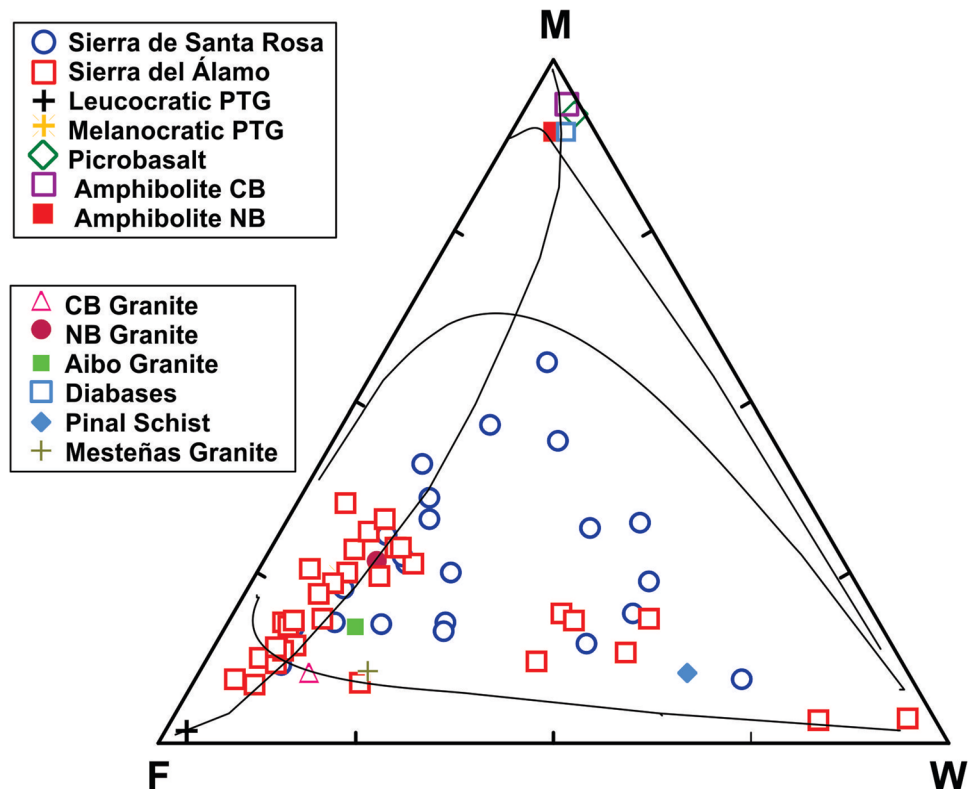
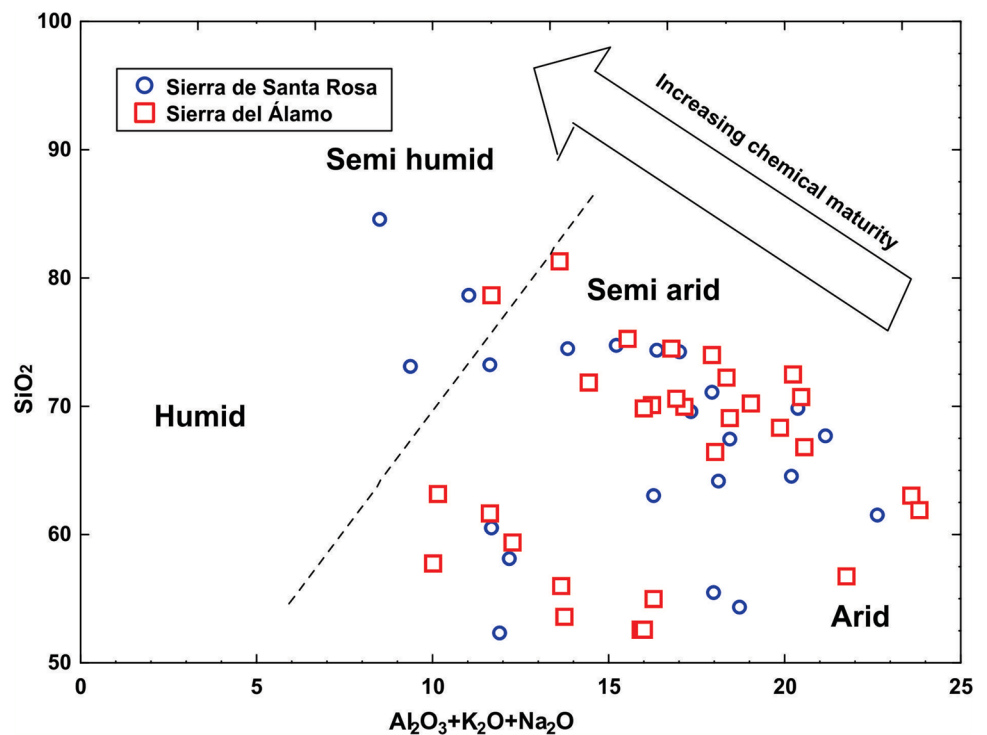


Fig. 14 $\text{Al}_2\text{O}_3 + \text{K}_2\text{O} + \text{Na}_2\text{O}\%$ vs $\text{SiO}_2\%$ bivariate diagram showing variations in paleolimatic condition for samples collected from Sierra del Álamo (Sierra del Álamo data from Galindo-Ruiz et al., 2025) and Sierra de Santa Rosa sections of Sierra de Santa Rosa Formation, Sonora, México (Suttner & Dutta, 1986)



et al., 1992; Dera et al., 2015; Hallam, 1984; Loope et al., 2004; Parrish, 1993). These warm and arid conditions allowed the formation of large ergs in the interior of the continent during the Jurassic, particularly in the southwest of North America, in the current states of Arizona and Utah, a

region known as the Colorado Plateau (Bilodeau & Keith, 1986; Dickinson & Gehrels, 2003, 2009), which could have acted as an important source of sediments during various periods of deposition.

The samples from Sierra de Santa Rosa and Sierra de Álamo sections have wide range of $\text{Al}_2\text{O}_3/\text{TiO}_2$ ratio values (11 to 31, $n=21$; 20 to 47, $n=31$; respectively). The twelve sandstone samples from the Sierra de Santa de Rosa section fall in the felsic composition whereas 9 samples show intermediate composition. However, majority of the samples from Sierra del Álamo section have felsic signature except three samples that show intermediate composition. This suggests that the contribution of intermediate source rocks was higher in the Sierra de Santa Rosa section than the Sierra del Álamo section. On the La/Th against Hf diagram (Fig. 15), most of the samples from both sections plot in the felsic source region, whereas few samples plot along the recycle source. On the La/Sc vs Co/Th diagram (Fig. 16), samples from both Sierra de Santa Rosa and Sierra del Álamo sections mostly fall in the felsic rocks trend. Most of the samples from both sections fall nearer to lueco and melanocratic Permo-Triassic granites (Arvizu et al., 2009), Pinal Schist (Madhavaraju et al., 2017), Carborca block granite, and North American block granite (Iriondo et al.,

2004) whereas some samples from the Sierra de Santa Rosa section plot nearer to Aibo Granite and Mestañas Granite. This diagram also indicates that the samples from Sierra de Santa Rosa section show significant variation in La/Sc ratio than the Sierra de Álamo section such variations may be occurred due to changes in the contribution of source rocks (i.e. Aibo Granite and Mestañas Granite). In addition, elemental ratios such as La/Sc and Th/Sc show wide variations between them in both sections whereas La/Co, Th/Co, and Cr/Th show limited variations (Supplementary Table 5). Overall, these elemental ratios in most of the samples from both sections fall within the range of felsic source rocks (Supplementary Table 6), while some samples have intermediate source rocks composition.

The detrital zircon U–Pb geochronological data of sandstone samples from the Sierra del Álamo and Sierra de Santa Rosa sections of the Sierra de Santa Rosa Formation exhibit that contemporaneous sedimentation that occurred during the Early Jurassic (Galindo-Ruiz et al., 2025; Hodges et al., 2017). The analyzed zircons from both sections yielded

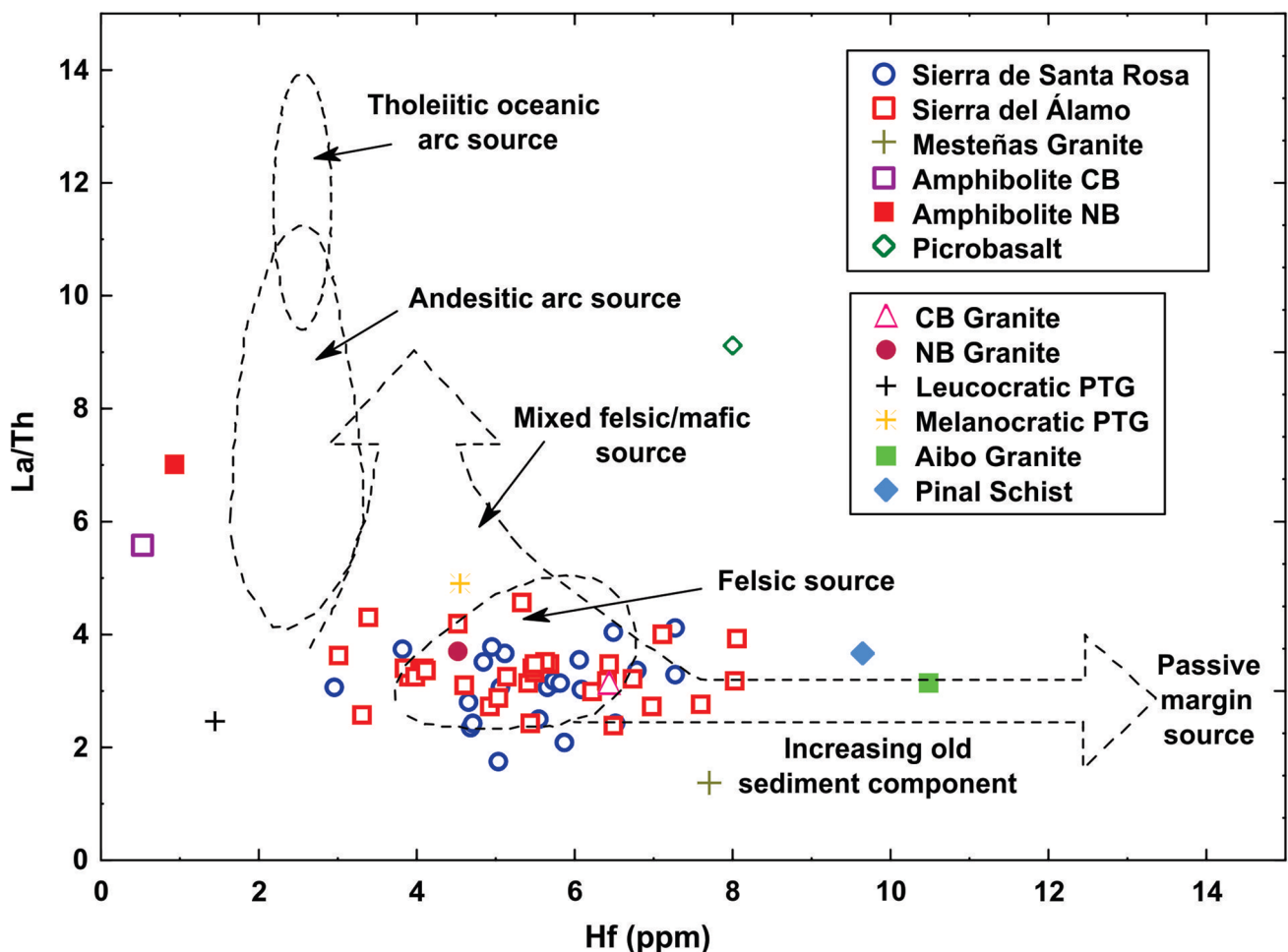


Fig. 15 La/Th vs Hf bivariate plot showing source rocks composition for samples collected from Sierra del Álamo (Sierra del Álamo data from Galindo-Ruiz et al., 2025) and Sierra de Santa Rosa sections of Sierra de Santa Rosa Formation, Sonora, México (Floyd & Leveridge, 1987)

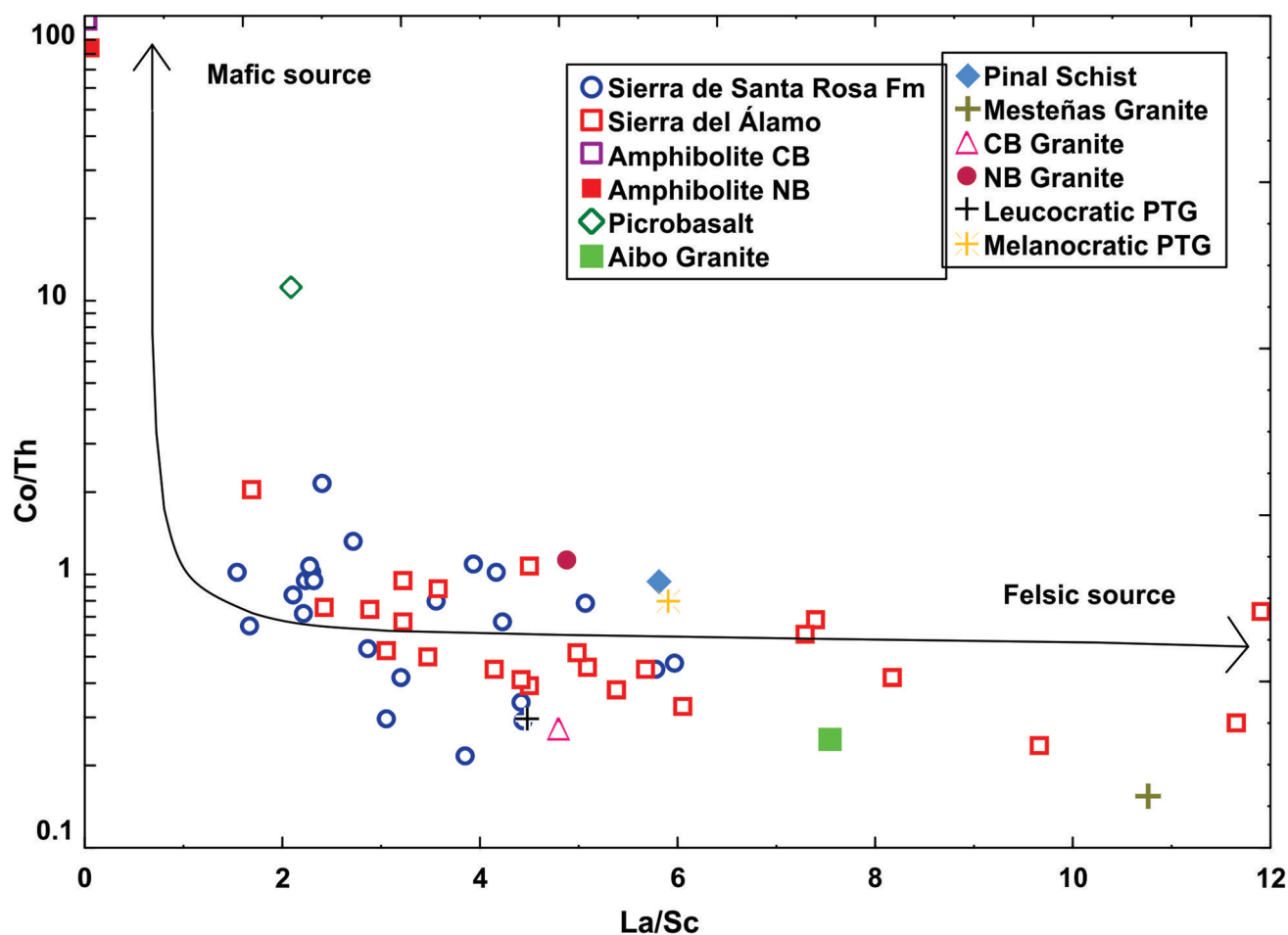


Fig. 16 Co/Th vs La/Sc bivariate plot showing source rock characteristics for samples collected from Sierra del Álamo and Sierra de Santa Rosa sections of Sierra de Santa Rosa Formation, Sonora, México (Sierra del Álamo data from Galindo-Ruiz et al., 2025)

Proterozoic, Late Triassic and Jurassic zircons as major populations, whereas Early Devonian and Middle Permian zircons present as minor populations. Meso and Paleoproterozoic group zircons are more or less similar in both sections, these zircons may be derived from the Precambrian basement rocks of southwestern USA and northwestern Mexico, and/or recycled detrital strata of Neoproterozoic to Triassic age of the same region (Hodges et al., 2017). However, the Neoproterozoic and Paleozoic zircon populations are more in the Sierra de Santa Rosa than in the Sierra del Álamo section (Galindo-Ruiz et al., 2025; Hodges et al., 2017). These zircon populations were mainly recycled from Paleozoic erg strata of the Colorado Plateau (Hodges et al., 2017).

Lower content of Permian zircon grains is present in both sections, however abundant Triassic zircon grains are present in the Sierra del Álamo section (Galindo-Ruiz et al., 2025) and it is absent in Sierra de Santa Rosa section (Hodges et al., 2017). The Permian and Triassic detrital zircon grains were mainly derived from the Permo-Triassic magmatic

arc whose outcrops are scarcely present in southern California and northwestern Sonora (Arvizu & Iriondo, 2015; Arvizu et al., 2009). Both sections have minor proportions of Jurassic zircon grains that might have been derived from the Jurassic continental volcanic arcs of southwestern North America (Busby-Spera et al., 1990; Hodges et al., 2017).

6 Conclusions

The chemical composition of the Lower Jurassic sandstones in the Sierra de Santa Rosa Formation, northern Sonora, Mexico is studied to infer the provenance and paleoweathering conditions of the basin within the Lower Jurassic paleogeographic framework. The concentrations of SiO_2 , Al_2O_3 and CaO are enriched in the sandstone samples than other oxides. Al_2O_3 show significant positive correlation with other oxides (Fe_2O_3 , MgO , K_2O , and TiO_2) supports their association with aluminosilicates. Among trace elements, Ba, Zr, Sr, and Rb have higher concentrations than other

trace elements. The CIA values for the sandstone range between 43 and 67 and when the samples plot in the A-CN-K diagram, they plot along the A-CN line suggesting an ideal weathering trend as well as low to moderate intensity of weathering in the source region. The chondrite-normalized REE patterns of the sandstone samples have light REE (LREE) enrichment, more or less flat heavy REE (HREE) patterns with negative Eu anomalies. The chondrite normalized REE patterns, $\text{Al}_2\text{O}_3/\text{TiO}_2$ ratio, trace and REE elemental ratios, and bivariate plots suggest that the sedimentary basin was contributed by majority of felsic rocks with minor contribution from intermediate source rocks.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s43217-026-00326-0>.

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Authors contribution Erik Ramírez-Montoya: Writing original draft, Writing, review and editing, Investigation, Methodology, Data Curation. Jayagopal Madhavaraju: Funding acquisition, Formal analysis, Supervision, Methodology, Validation, Writing original draft. Writing, review and editing. John S. Armstrong-Altrin: Methodology, Validation, Writing, review and editing. Raul Alfredo Pinelli—Castillo: Formal analysis, review and editing. Ofelia Perez Orvizu: Formal analysis, Data Curation.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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